

Error Annotation for English Language Learning

Annotation Scheme v.1.1

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1 Introduction

This project explores how we can estimate an English learner’s proficiency level based on the types of errors they make when writing. An error is a mistake in spelling, punctuation, grammar, or word choice. We are most interested in grammatical and word choice error types since their presence can indicate higher or lower proficiency levels.

1.1 Annotation goal

Our goal is to annotate the types of errors present in our chosen dataset and assign a proficiency level to each text sample. We wanted to find a more concrete way to measure proficiency using patterns in the errors themselves, so we take into consideration how many errors a learner makes and what kinds of errors they are. In this way, we will have a more consistent way to assess writing ability. This will allow for further development of systems that can aid in English language learning. It could also aid in the development of automated English language error correction systems.

1.2 Summary of task framing

This annotation task has two parts:

Task 1 is to annotate different types of errors in written essay prompt responses by English language learners. We have identified some common error types that are present in the data and have created labels for them. Some of the labels will require a dependency – we have specified such labels below.

Task 2 is to assign a proficiency level as an annotation on the entire text. The proficiency levels represented in this dataset use the Common European Framework of Reference for Languages (CEFR). From least proficient to most proficient, these levels include A1, A2, B1, B2, C1, and C2. We have condensed the proficiency levels into Basic, Intermediate, and Advanced.

In the context of these tasks, an error is a deviation from standard grammatical, lexical, or discourse conventions of English that affects clarity, fluency, or correctness in communication.

For each data sample, perform Task 1, then move on to Task 2.

1.3 Corpus description

COREFL (CORpus of English as a Foreign Language) is a comprehensive English learner corpus that encompasses both written and spoken data from 2,342 participants, totaling approximately 530,392 words. Designed following the principles of the CEDEL2 corpus, COREFL is designed to facilitate research into second language acquisition. The corpus includes learner subcorpora featuring L1 Spanish–L2 English and L1 German–L2 English data, along with native control subcorpora that consists of Spanish and English speakers. We will be using the written data from L1 Spanish–L2 English participants. Detailed metadata accompany each text file, covering aspects such as participant identifiers, placement test scores, CEFR-aligned proficiency levels, demographic and linguistic background information (including L1, parental languages, and language exposure), as well as task-specific variables like task title, medium, time taken, and resources used. COREFL is distributed under a Creative Commons CC BY-NC-ND 3.0 ES license.

Lozano, C., Díaz-Negrillo, A., & Callies, M. (2020). Designing and compiling a learner corpus of written and spoken narratives: COREFL. In C. Bongartz & J. Torregrossa (Eds.), *What's in a Narrative? Variation in Story-Telling at the Interface between Language and Literacy* (pp. 21-46). Peter Lang. <https://doi.org/10.3726/978-3-653-05182-7>

2 Label Studio Software

All annotations for this task should be done via (Label Studio).

2.1 Label Studio Resources

- [Install Label Studio](#)
- [Label Studio Documentation](#)

2.2 Technical Setup

Install Label Studio. It can be used locally for this task. The [configuration file](#) is set up for use within Label Studio.

2.2.1 File Organization

- Here is the [main Google Drive folder](#)
- All json files are stored in this [Google Drive folder](#)
- Each annotator will work on one file labeled `annotation_group[i]`, where `[i]` is a number
- Save annotation files like `EllErrorAnnotation_Lastname_Firstname_group[i]`

3 Britishisms

These are words and phrases that are used by people from the UK. They are not errors and should not be annotated as such.

This section contains a list of some Britishisms (along with their definitions) that may be present in the data. There may be other Britishisms that are not included in this list. In those cases, additional research may be required by the annotator to ensure validity of the suspected Britishism.

- pushchair, baby cart - a stroller or baby carriage
- at the weekend - on the weekend
- colour - color
- realise - realize

4 Error types

These are the different error types. Assign labels to tokens or token sequences in each sample based on the instructions given for each label. Follow the error type label hierarchy - each token or token sequence will get one label at most, and the order in which to assign the labels is the order they are listed in this document. If the error is that a token is missing,

label the empty space where the token should be. For any example, if the errors make the meaning incomprehensible, label the main error types and assign a proficiency level. We have provided examples for each label within this document, and we have additional Label Studio examples [here](#).

Symbols used:

- $\emptyset$ - represents a missing word
- # - denotes an example containing the error

4.1 Preposition error

Incorrect usage of a preposition, such the presence of a preposition where there should not be one, the absence of a preposition where there should be one, or the ungrammatical usage of a different preposition.

In the next morning, Josh took a walk.
The next morning, Josh took a walk

Josh goes $\emptyset$ the beach.
Josh goes to the beach.

Josh walked **in** the sidewalk.
Josh walked on the sidewalk.

Negative examples (non-errors):
Josh walked **in** the street.

4.2 Article error

Incorrect usage of an article, including the presence of an article where there should not be one, the absence of an article where there should be one, or a disagreement between the article and the noun.

At **the** night when Charles and his dog were sleeping
At night when Charles and his dog were sleeping

Charles went to $\emptyset$ store.
Charles went to the store.

A dogs ran.
The dogs ran.

Negative examples (non-errors):
A dog is a wonderful pet.

4.3 Pronoun case error

Incorrect usage of pronoun case.

Hannah gave the gift to **he**.
Hannah gave the gift to him.

Negative examples (non-errors):
He is taller than **I**.
Her dog and **she** walk to the park.

4.4 Subject-verb agreement error

The subject and the verb do not agree in number. Before labeling this error type, read the entire sentence to ensure it isn't actually a verb tense error.

Charles and his dog **was** sleeping.
Charles and his dog were sleeping.

Negative examples (non-errors):
The team **is** winning the match.

4.5 Verb tense error

Incorrect usage of verb tense. Includes all non subject-verb agreement errors related to tense. Tag the entire verb sequence. Before labeling this error type, read the entire sentence to ensure it isn't actually a subject-verb agreement error.

Yesterday the frog **escape**.
Yesterday the frog escaped.

The girl decided to **found** it.
The girl decided to find it.

Yesterday Jim **is sleep** at 9 o'clock.
Yesterday Jim was sleeping at 9 o'clock.

He suggested **to go** home.
He suggested going home.

Negative examples (non-errors):
She **has gone** to the store.
They **have not been eating** the cereal.

4.6 Subject or expletive omission error

A false pro-drop rule is applied (subject pronoun or named entity is omitted), or an expletive (e.g., there, it) is not present.

$\emptyset$ Pushed the boy into the water.
Lucy pushed the boy into the water.
Push the boy into the water! [command]

$\emptyset$ Was a deer.
There was a deer.
It was a deer.

I can't find $\emptyset$.
I can't find it.

Negative examples (non-errors):
Seems like a good idea.

4.7 Subject-verb order error

Ungrammatical SVO order. Label the main verb or verb phrase.

The bus stop **passes** Charles. [where Charles passes the bus stop]
Charles passes the bus stop.

Jumps Charles
Charles jumps

Negative examples (non-errors):
Rarely **does** she go out at night.
Did you hear that?

4.8 Nominalization error

An adjective ungrammatically used as a noun.

I saw a homeless $\emptyset$.

I saw a homeless person.

Negative examples (non-errors):

The homeless need shelter.

4.9 Number agreement error

Incorrect singular/plural noun forms or mismatches with determiners. Mismatches between adjectives and countable/uncountable nouns.

Hannah has five **apple**.

Hannah has five apples.

The **informations** are useful.

The information is useful.

Marni ate too **much** peas.

Marni ate too many peas.

Negative examples (non-errors):

The news **is** shocking.

That was a lot of smiling.

4.10 Other

All errors that have not been covered by labels pertaining to specific error types or pre-annotated errors, for example:

I **have** 30 years.

I am 30 years old.

I **did** a mistake.

I made a mistake.

He was in **him** bedroom.

He was in his bedroom.

Negative examples (non-errors):

I had a good time.

Nice weather today, innit?

4.11 Pre-annotated errors

There are some errors that have been pre-annotated. In the case that an error is identified as one of the errors listed below but has not been assigned a label, you do not need to assign it a label, although you can if you want to. In the case that one of the labels below has been incorrectly assigned to a word or series of words, you do not need to correct it. The number of correct pre-annotations is much higher than the number of incorrect pre-annotations, and the priorities should fall to labeling grammatical errors.

4.11.1 Spelling errors

Incorrect spelling of a word.

The fox was hiden from sight.
The fox was hidden from sight.

Molly throughs me the ball.
Molly throws me the ball.

Negative examples (non-errors):
Blue is my **favourite** colour.
Nancie is my best friend.

4.11.2 Punctuation

Incorrect usage of punctuation marks, or absence of punctuation marks where they should be.

We walked up to the manager, he smiled at us.
We walked up to the manager. He smiled at us.

–What do you mean? –Sarah asked–. There’s nothing there.
”What do you mean?” Sarah asked. ”There’s nothing there.”

Negative examples (non-errors):
I tried to call them; however, they didn’t answer.

4.11.3 Capitalization errors

Incorrect usage of capital letters, or absence of capital letters where they should be.

the sky is blue.
The sky is blue.

My dog is named **henry**.
My dog is named Henry.

I lost my **Water Bottle**.
I lost my water bottle.

Negative examples (non-errors):
We go to the library on **Mondays**. Valentine's Day is in **February**.

5 Proficiency level

Assign a proficiency level to each text sample based on the overall complexity of the errors and the fluency of the writing. The examples given for each proficiency level listed below are all positive examples – that is, they are examples of texts that fit the proficiency level. The examples in each category can be compared with the examples in the other categories to determine how to annotate each text sample.

5.1 Basic

- Frequent, fundamental grammar errors, including basic syntax mistakes (e.g., missing subjects, incorrect word order)
- Errors in high-frequency structures, such as simple verb tenses, articles, prepositions, and plurality
- Many direct translations from the learner's first language, leading to unnatural phrasing
- Simple sentence structures with limited vocabulary and frequent word choice errors
- May struggle with basic spelling and punctuation
- Limited use of cohesive devices

One day,...in mornig a boy is look at the frog wif her bog Rocky. The go sleep
tha frog escape in morning the boy see her frog they go find to the frest and shout

This video tell us a story about a poor man who find a baby on the street. He
doesn't know what can he does and decide give him to a woman who walks down

street. She doesn't want the baby and she return it. After that the man see other man walking down and decide to give the baby to him and then he run very fast. But this man doesn't like him and his runs after the poor man too. [...]

5.2 Intermediate

- Some errors in verb tense, prepositions, articles, and subject-verb agreement, but the meaning is generally clear
- Occasional unnatural phrasing due to incorrect collocations or influence from another language
- More varied sentence structures with some complexity, but occasional errors in clause formation
- Vocabulary is broader than at the basic level, but word choice mistakes may still be present
- Spelling and punctuation errors are less frequent but may still appear

One day, there was a boy in his bedroom with its dog, called Clue and a new frog. It was very late so the boy called Maria decided to sleep. While he was sleeping, the frog escaped from the vase. In the morning, Mario saw that the frog wasn't there, so he decided to look for it. He went to the forest and he started to shout, he wanted found its frog! He looked behind the trees, a shore_ and when he was on a shorea deer pushed him and he fell in a river. [...]

A man is walking in the street, he is moving away of the place where rocks from the houses are falling down on to the floor. While he is smoking in a fun way, he finds a baby laying on the floor, that baby seems to be feeling blue because he is crying a lot. The man takes the baby with him and sees a powerful woman carrying her baby with him. So he goes to her and places the found baby with the other baby. When the woman realise what is happening she starts shouting at him and he has to take the baby back with him. [...]

5.3 Advanced

- Errors are minor and do not significantly impact readability or fluency
- Sentence structures are varied and complex, with correct use of subordinate clauses, connectors, and more advanced grammar structures

- Errors mostly occur in more subtle aspects of grammar, such as collocations, idioms, or advanced verb forms (e.g., conditionals, subjunctive)
- Strong vocabulary usage with occasional minor word choice issues, but no major meaning distortions
- Spelling and punctuation are mostly correct

Once upon a time, there was a little kid who lived happily with his dog and his frog. Unfortunately, the frog didn't feel as comfortable living there as the dog did, since it profoundly missed its family and its long routs throughout the swaps. One day, the frog decided to leave, and after getting its little luggage ready, it parted. The kid and the dog were worried sick and decided to go look for it. They now felt lonely without it, it was an important part of their lives. [...]

From afar comes a man, actually, he is Charles Chaplin, he wears a scruffy suit, a hat and big shoes. This bizarre outfit gives him a funny air. He has a cigarette between his lips and when he blows the smoke comes as if he was a steam locomotive. He walks like nothing around him matters, like he is oblivious of his surrounding. He passes below an open window, from which a man appears and throw something white, perhaps dust or flour, who knows. [...]

5.4 Measurement

To quantify proficiency levels, we use a **weighted error scoring** system. Different error types contribute different point values based on their severity:

Each error type is assigned a weight based on its severity:

Error Type	Weight
Subject omission	4
Subject-verb order	4
Tense/conjugation error	3
Subject-verb agreement	3
Preposition error	2
Article error	2
Pronoun case error	2
Nominalization error	2
Number agreement error	2
Other (e.g., unnatural phrasing)	1

Table 1: Error Weighting by Severity

$$ErrorDensity = \left(\frac{\sum (Errors \times Weight)}{TotalWords} \right) \times 100$$

1. If the text contains **3 or more** errors from the highest-weighted category (weight 4), it is automatically classified as Basic.
2. Otherwise, assign proficiency based on error density:
 - **Basic:** Error Density $\geq 30\%$
 - **Intermediate:** $15\% \leq \text{Error Density} < 30\%$
 - **Advanced:** Error Density $< 15\%$
3. For longer paragraphs (over 200 words), error density thresholds are adjusted by -5 percentage point to account for more natural variance in longer texts.